

Question [SET A]

Design the **Uber** class to generate the correct output from the driver code provided below:

Note : Fare calculation

- Bike : baseFare = 30Tk and perKm = 20;
- Car : baseFare = 80Tk and perKm = 35;
- CNG : baseFare = 50Tk and perKm = 25;

Driver Code	Output
<pre>public class UberTester { public static void main(String[] args) { Uber u1 = new Uber(); u1.showDetails(); System.out.println("1====="); u1.updateRide("Car", 10); u1.usertype = "Premium"; System.out.println("2====="); u1.showDetails(); System.out.println(u1.addPromo("FIFA20")); u1.showDetails(); Uber u2 = new Uber(); u2.updateRide("Bike", 8); u2.usertype = "Premium"; System.out.println("3====="); System.out.println(u2.addPromo("FIFA10")); u2.showDetails(); System.out.println("4====="); Uber u3 = new Uber(); u3.updateRide("CNG", 8); System.out.println(u3.addPromo("FIFA20")); u3.showDetails(); } }</pre>	<pre>Vehicle Type : Not select User Type : Regular Distance : 0.0 km Final Fare : 0.0 Tk 1=====</pre> <pre>2=====</pre> <pre>Vehicle Type : Car User Type : Premium Distance : 10.0 km Final Fare : 430.0 Tk Promo added with a 20% discount! Vehicle Type : Car User Type : Premium Distance : 10.0 km Final Fare : 344.0 Tk 3=====</pre> <pre>Invalid promo Vehicle Type : Bike User Type : Premium Distance : 8.0 km Final Fare : 190.0 Tk 4=====</pre> <pre>Promo Not Applied! Vehicle Type : CNG User Type : Regular Distance : 8.0 km Final Fare : 250.0 Tk</pre>

Question [SET B]

Design the **Pathao** class to generate the correct output from the driver code provided below:

Note : Fare calculation

- Bike : baseFare = 30Tk and perKm = 22;
- Car : baseFare = 80Tk and perKm = 40;
- CNG : baseFare = 50Tk and perKm = 30;

Driver Code	Output
<pre>public class PathaoTester { public static void main(String[] args) { Pathao p1 = new Pathao(); p1.distance = 10; p1.setRideInfo("Car", "Premium"); System.out.println("1====="); System.out.println(p1.applyPromo("FIFA15")); System.out.println("2====="); p1.showRideDetails(); System.out.println("3====="); Pathao p2 = new Pathao(); p2.showRideDetails(); p2.distance = 8; p2.setRideInfo("Bike", "Regular"); System.out.println("4====="); System.out.println(p2.applyPromo("FIFA15")); p2.showRideDetails(); Pathao p3 = new Pathao(); p3.distance = 8; p3.setRideInfo("CNG", "Premium"); System.out.println("5====="); System.out.println(p3.applyPromo("FIFA20")); System.out.println("6====="); p3.showRideDetails(); } }</pre>	<pre>1===== Promo added with a 15% discount! 2===== Vehicle Type : Car User Type : Premium Distance : 10.0 km Final Fare : 408.0 Tk 3===== Vehicle Type : Bike User Type : Default Distance : 0.0 km Final Fare : 0.0 Tk 4===== Promo Not Applied! Vehicle Type : Bike User Type : Regular Distance : 8.0 km Final Fare : 206.0 Tk 5===== Invalid promo 6===== Vehicle Type : CNG User Type : Premium Distance : 8.0 km Final Fare : 290.0 Tk</pre>

Solutions

SET A	SET B
<pre> public class Uber { public String vehicleType = "Not select"; public String usertype = "Regular"; public double distance; public double fare; // Update vehicle and distance public void updateRide(String vehicleType, double distance) { this.vehicleType = vehicleType; this.distance = distance; double baseFare = 0; double perKm = 0; if (vehicleType.equals("Bike")) { baseFare = 30; perKm = 20; } else if (vehicleType.equals("Car")) { baseFare = 80; perKm = 35; } else if (vehicleType.equals("CNG")) { baseFare = 50; perKm = 25; } fare = baseFare + (distance * perKm); } // Promo method public String addPromo(String promo) { if (usertype.equals("Premium") && promo.equals("FIFA20")) { fare = fare * 0.80; // Extra 20% discount return "Promo added with a 20% discount!"; } if (!promo.equals("FIFA20")) { return "Invalid promo"; } return "Promo Not Applied!"; } // Show Details public void showDetails() { System.out.println("Vehicle Type : " + vehicleType); System.out.println("User Type : " + usertype); System.out.println("Distance : " + distance + " km"); System.out.println("Final Fare : " + fare + " Tk"); } } </pre>	<pre> public class Pathao { public String vehicleType = "Bike"; public String usertype ="Default"; public double distance; public double fare; // Update vehicle and distance public void setRideInfo(String vehicleType, String type) { this.vehicleType = vehicleType; this.usertype = type; double baseFare = 0; double perKm = 0; if (vehicleType.equals("Bike")) { baseFare = 30; perKm = 22; } else if (vehicleType.equals("Car")) { baseFare = 80; perKm = 40; } else if (vehicleType.equals("CNG")) { baseFare = 50; perKm = 30; } fare = baseFare + (distance * perKm); } // Promo method public String applyPromo(String promo) { if (usertype.equals("Premium") && promo.equals("FIFA15")) { fare = fare * 0.85; return "Promo added with a 15% discount!"; } if (!promo.equals("FIFA15")) { return "Invalid promo"; } return "Promo Not Applied!"; } // Show Details public void showRideDetails() { System.out.println("Vehicle Type : " + vehicleType); System.out.println("User Type : " + usertype); System.out.println("Distance : " + distance + " km"); System.out.println("Final Fare : " + fare + " Tk"); } } </pre>

Rubric for Set A and Set B

<u>SL</u>	Points to Meet	Marks(10)
1.	Correct class creation (Uber/Pathao)	1
2.	Proper attributes /instance variable used	1
3.	Proper updateRide((String vehicleType, double distance) / setRideInfo(String vehicleType, String type) method + value update	1+ 1
4.	Proper fare calculate	2
5.	Promo System (validity check + return)	1+1
6	showDetails() / showRideDetails()	2